Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions

Made with GAMMA

Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Customer and purchase details

37

Missing Values

Review ratings imputed

Key Data Points

- Demographics: Age, Gender, Location
- Purchase details: Category, Amount, Season
- Behavior: Discounts, Reviews, Shipping

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Python Data Preparation

01

Data Loading

Imported using pandas

02

Exploration

Checked structure and statistics

03

Cleaning

Handled missing values

04

Standardization

Renamed columns to snake case

05

Feature Engineering

Created age groups and frequency metrics

06

Database Load

Connected to PostgreSQL

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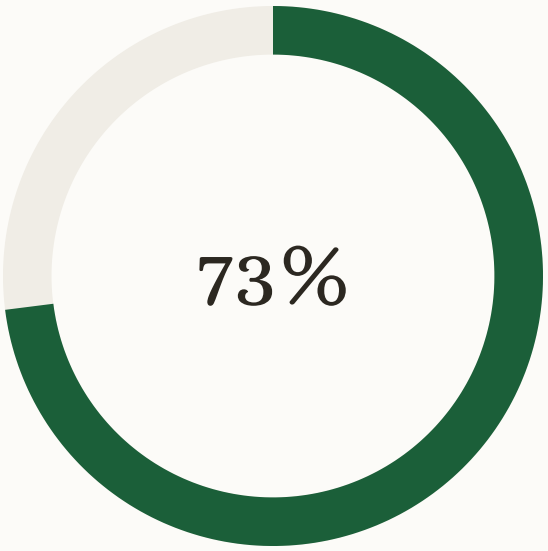
SQL Analysis: Revenue Insights

Revenue by Gender

- **Male:** $157,890
- **Female:** $75,191

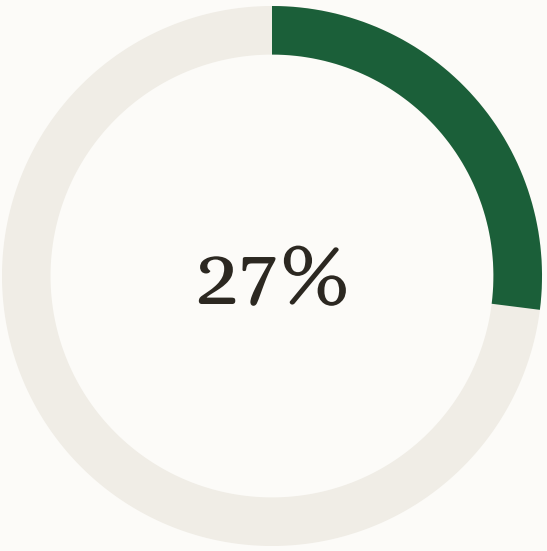
Shipping Comparison

- **Standard:** $58.46 avg
- **Express:** $60.48 avg



Non-Subscribers

2,847 customers



Subscribers

1,053 customers

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Product Performance

Gloves

3.86 avg rating

Sandals

3.84 avg rating

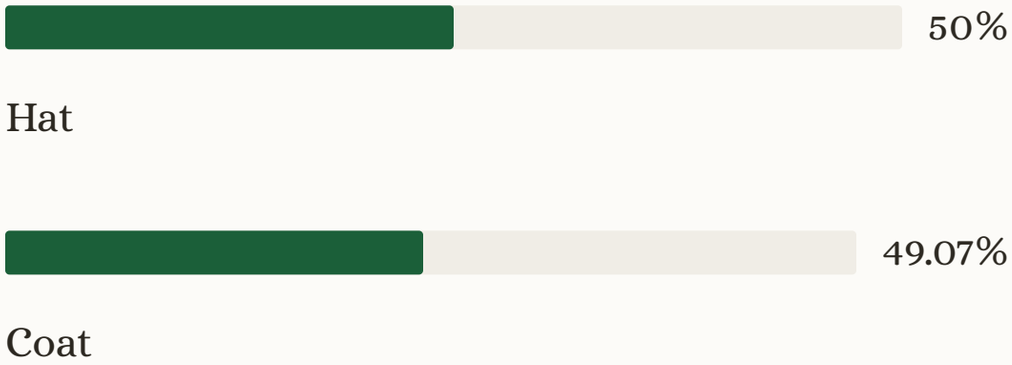
Boots

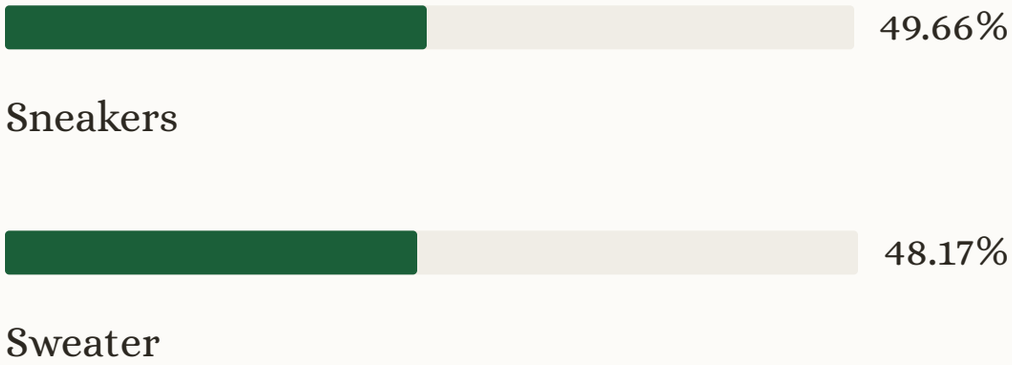
3.82 avg rating

Hat

3.80 avg rating

Top Discount Products





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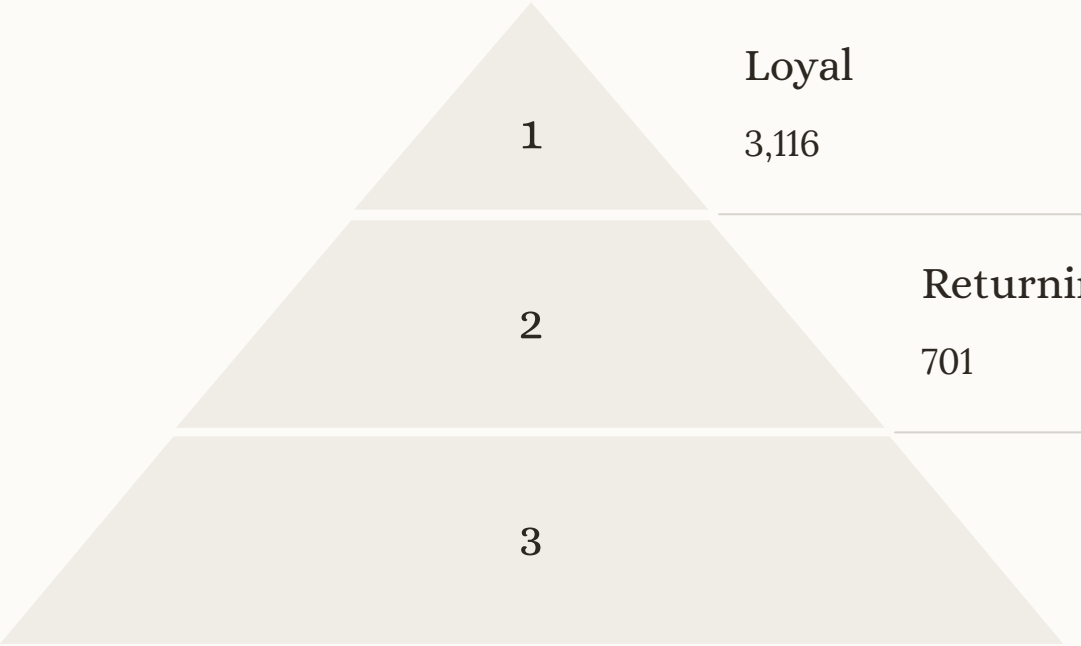
Customer Segmentation



Loyal

3,116

Returning

701

New

83

Repeat Buyers

958 customers with >5 purchases

Subscription Link

Repeat buyers more likely to subscribe

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Revenue by Age Group

Young Adult

$62,143

Middle-aged

$59,197

Adult

$55,978

Senior

$55,763

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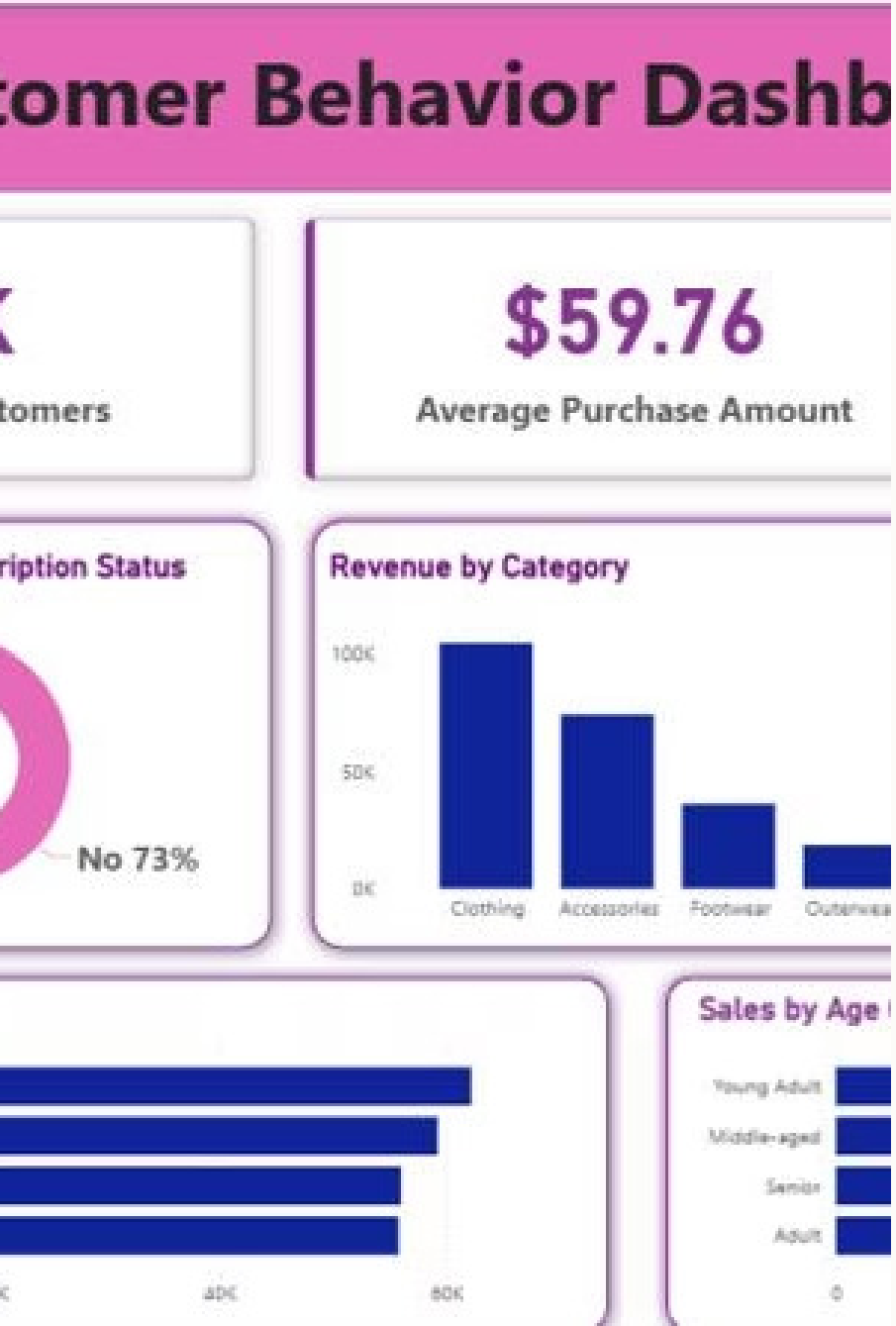

Power BI Dashboard

KPIs

- 3,900 customers
- $59.76 avg purchase
- 3.75 avg rating

Filters

- Subscription status
- Gender, category
- Shipping type

Visualizations

- Revenue by category
- Sales by age group
- Subscription status

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Key Findings

Gender Gap

Male customers generate 2.1x more revenue than female customers

Discount Strategy

50% of hat purchases use discounts—balance sales with margin control

Loyalty Focus

80% of customers are loyal—reward repeat buyers to boost retention

Age Groups

Young adults drive highest revenue—target marketing to this segment

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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